

Civilization Simulation MMO

Updated Game Design Document (GDD)

Core premise: The player is the head of state of an entire civilization. The player controls the state; the simulation controls the civilization.

1. High Concept

A persistent, multiplayer, terminal-based grand strategy and civilization simulation where each player governs an entire civilization.

The player is the **Head of State**. They control government policy, laws, taxes, public spending, infrastructure, economic strategy, foreign policy, military strategy, research priorities, industrial development, education, diplomacy, intelligence, and long-term national objectives.

The player does not directly control the population. Citizens, businesses, politicians, generals, governors, scientists, merchants, organizations, and foreign powers are simulated actors with their own interests.

Design philosophy: The player controls the state. The simulation controls the civilization.

2. Design Pillars

Civilization, not character: The player governs a civilization rather than inhabiting an individual role.

Everything is connected: Resources, people, technology, institutions, infrastructure, markets, politics, diplomacy, and war influence one another.

Resources become capabilities: A resource is not a simple unlock. Its value depends on industries, institutions, infrastructure, knowledge, labor, and capital.

Consequences matter: Decisions propagate through the simulation and can create unintended outcomes.

3. The Player

Can: enact laws; issue policies; establish institutions; allocate budgets; set taxes; subsidize industries; establish trade agreements; declare wars; negotiate treaties; appoint officials; reorganize administration; set research priorities; order military campaigns; build infrastructure; regulate markets; fund education and healthcare; control immigration.

Cannot directly: command every citizen; instantly create industries; guarantee political support; make businesses profitable; force scientific discoveries; eliminate corruption instantly; guarantee military obedience; know everything happening inside the country.

The player governs through systems and institutions rather than direct unit-by-unit control.

4. Civilization Model

Core layers: Geography → Resources → Population → Production → Industries → Markets → Economy → Institutions → State → Politics → Diplomacy → Military.

Technology interacts with all layers rather than existing as a simple technology tree.

5. Geography

Regions contain terrain, climate, fertility, water, mineral deposits, forests, wildlife, strategic resources, coastline, rivers, harbors, elevation, population capacity, infrastructure, settlements, cities, and political ownership.

Resources are continuous rather than binary. A region may have high horse population, high breeding quality, strong pasture availability, moderate transport infrastructure, low saddlery production, and high military demand.

6. Resources

Resource categories include biological resources (horses, cattle, grain, timber, cotton), minerals (iron, copper, coal, silver, gold), energy resources (wood, coal, oil, gas, hydro potential), and strategic assets such as fertile land, freshwater, rivers, harbors, and passes.

Important resource properties include availability, abundance, quality, accessibility, production, processing, transportability, storage, demand, price, strategic importance, known uses, and developed uses.

Having a resource does not mean having the capability associated with it.

7. Resource → Industry → Capability

One of the game's fundamental systems is: Resource → Processing → Industry → Institution → Infrastructure → Capability.

For example, horses can develop into horse breeding, animal husbandry, breeding institutions, stud farms, roads, and stables, eventually creating a national horse industry.

8. Example: Horses

Agriculture: plowing, hauling, harvesting, transport.

Transportation: carts, carriages, freight, passenger transport.

Military: cavalry, mounted scouts, horse artillery, logistics, mounted infantry.

Communications: couriers, postal routes, messenger networks.

Economy: saddles, harnesses, stables, blacksmiths, veterinary services, breeding.

Society: ownership classes, racing, prestige, sports, transportation professions.

Government: postal services, military remount departments, breeding programs, veterinary institutions, road networks.

Thus, a resource becomes a civilizational capability rather than a binary technology-tree unlock.

9. Technology

Technology is divided into four layers: **Knowledge** (mathematics, metallurgy, biology, navigation, medicine); **Techniques** (steelmaking, horse breeding, irrigation, steam engines); **Institutions** (universities, banks, academies, postal services); and **Infrastructure** (roads, railways, factories, laboratories, power grids).

Capability = Knowledge + Technique + Institutions + Infrastructure + Resources + Skilled Population + Capital + Administrative Capacity.

Discovering something does not automatically mean the civilization can use it effectively.

10. Technology Diffusion

Knowledge can spread through trade, universities, espionage, migration, conquest, diplomacy, books, scientific exchange, corporations, missionaries, captured equipment, and foreign investment.

A civilization may know that a technology exists while lacking the industrial base, institutions, or skills required to reproduce it.

11. Population

Population is simulated in aggregate. Groups have location, age distribution, wealth, profession, education, culture, political ideology, religion, satisfaction, employment, income, living conditions, migration preferences, and political affiliation.

Example groups include farmers, industrial workers, engineers, merchants, soldiers, bureaucrats, scientists, aristocrats, students, artisans, clergy, and unemployed populations.

12. Population as Actors

Population groups react to taxes, prices, unemployment, wages, wars, laws, education, economic opportunity, migration, and political conditions.

Possible outcomes include protests, strikes, migration, political movements, rebellion, entrepreneurship, recruitment, government support, and radicalization.

13. Businesses

Businesses produce goods, employ workers, buy inputs, sell products, compete, invest, fail, expand, lobby, evade taxes, and innovate.

Behavior depends on expected profit, risk, capital, demand, taxes, regulation, labor costs, resource prices, infrastructure, political stability, and technology.

14. Industries

Industries emerge from combinations of resources, labor, capital, knowledge, technology, infrastructure, and demand.

The player cannot simply click 'Build Automobile Industry.' The required economic ecosystem must exist. The state can accelerate development through subsidies, tariffs, state-owned companies, education, infrastructure, research, investment, and procurement.

15. Economy

The economy contains production, consumption, prices, wages, employment, capital, investment, banking, credit, debt, trade, taxation, government spending, inflation, and currency.

Markets exist at local, regional, national, and international levels. Prices depend on supply, demand, transport cost, taxes, tariffs, market power, storage, and expectations.

16. Infrastructure

Transportation includes roads, bridges, canals, railways, ports, and airports. Communication includes postal routes, telegraph, telephone, radio, and internet. Energy includes coal networks, oil pipelines, electrical grids, and nuclear infrastructure. Social infrastructure includes schools, universities, hospitals, and sanitation.

Infrastructure determines how effectively a civilization can exploit resources and knowledge.

17. Administrative Capacity

Administrative capacity limits the player's practical power. It depends on bureaucracy, education, communication, transportation, centralization, institutions, corruption, political legitimacy, and government organization.

Example: a 20% official tax rate may produce only 11% actual collection because of corruption, local evasion, and weak administration.

18. Information

The player does not have perfect information. Information may be unknown, rumored, public, reported, estimated, verified, classified, or obtained through intelligence.

Players should sometimes need intelligence, bureaucracy, communication infrastructure, or investigation to understand what is really happening.

19. Politics

Political groups emerge from society: industrialists, workers, farmers, military, aristocracy, liberals, conservatives, socialists, nationalists, regionalists, religious movements, and others.

Political power depends on wealth, population, organization, institutions, public opinion, military support, media influence, and economic importance.

20. Government and Laws

Government structures may include monarchy, republic, parliamentary government, presidential government, dictatorship, oligarchy, federation, confederation, and empire.

Government types determine how decisions are made and constrained rather than simply providing static bonuses.

Laws can cover taxation, labor, property, immigration, education, military service, business regulation, trade, censorship, healthcare, welfare, and land ownership.

21. Elections and Legitimacy

Depending on the political system, the player may face elections, parliamentary opposition, coups, revolutions, protests, strikes, and separatism.

The player remains the head of state, but their ability to remain in power depends on the political system.

22. Diplomacy and Trade

Diplomatic relationships depend on trade, ideology, borders, military power, historical relations, resources, alliances, territorial claims, and economic dependence.

Trade routes have volume, capacity, transportation costs, tariffs, risks, blockades, piracy, infrastructure requirements, and political dependencies.

23. Military

Military power depends on manpower, food, weapons, ammunition, logistics, transportation, officers, training, industry, technology, and money.

Military doctrine emerges from technology, resources, geography, institutions, historical experience, officers, and training.

24. War

Combat depends on logistics, terrain, morale, leadership, doctrine, intelligence, equipment, supply, infrastructure, weather, industrial production, population, and strategic objectives rather than a single combat-power number.

25. AI Civilizations

AI civilizations evaluate their economy, identify threats, pursue strategic objectives, form alliances, develop industries, manipulate diplomacy, react to player actions, make mistakes, and have different ideologies and risk tolerances.

The goal is believable behavior, not perfect optimization.

26. Civilization Identity

Identity emerges from history rather than static civilization bonuses. A civilization may gradually become centralized, industrial, militaristic, urbanized, educated, nationalistic, maritime, or financially powerful.

Another may become decentralized, agricultural, commercial, federal, rural, religiously conservative, or diplomatically oriented.

27. Events and Causal Simulation

Events should emerge from the simulation: food shortages, banking crises, industrial strikes, scandals, coups, scientific breakthroughs, disease outbreaks, commodity booms, resource discoveries, corruption scandals, border crises, revolutions, and migration waves.

Players should be able to investigate causes. Example: war → coal imports interrupted → coal prices rise → steel production falls → railway maintenance declines → transport capacity falls → food distribution problems → urban food prices rise → unrest → political opposition.

28. World Time

Initial target: 1 real hour = 1 game month. This should remain configurable during development.

The simulation uses timestamp-based progression and catch-up rather than assuming ticks always execute exactly on time.

29. Simulation Tick

Typical pipeline: Advance Time → Geography/Environment → Resource Production → Population → Businesses → Markets → Economy → Infrastructure → Institutions → Politics → Diplomacy → Military → Technology → AI Decisions → Events → Notifications → Persist State.

30. Simulation Architecture

The simulation must be independent of the UI.

Suggested structure: /apps/web; /packages/simulation; /packages/economy; /packages/population; /packages/politics; /packages/military; /packages/diplomacy; /packages/technology; /packages/ai; /packages/shared.

The frontend should never contain core simulation logic.

31. Technology Stack

Frontend: Astro, React, TypeScript.

Backend: Supabase, PostgreSQL, Supabase Auth, Supabase Realtime, Supabase Edge Functions.

Hosting: Netlify.

Simulation: TypeScript initially, with the option to move computationally expensive systems to another runtime later.

32. Database Philosophy

PostgreSQL is the authoritative source of persistent world state.

Core entities include worlds, civilizations, regions, cities, population groups, businesses, industries, resources, resource deposits, markets, goods, infrastructure, institutions, technologies, knowledge, laws, political groups, political parties, governments, armies, navies, wars, treaties, trade routes, events, decisions, and notifications.

33. Player Decisions and Projects

Important player actions become decisions/orders rather than direct database mutations.

Large actions become projects with cost, duration, labor, materials, institutions, prerequisites, political support, risks, expected outcomes, and actual outcomes.

Examples: railways, universities, factories, naval bases, postal systems, central banks, electrification, canals, and colonization.

34. Institutions

Institutions are organizations capable of preserving knowledge and making complex systems function: parliament, central bank, university, postal service, military academy, railway administration, tax authority, intelligence service, public health service, stock exchange, scientific academy.

Institutions have quality characteristics such as funding, personnel, prestige, output, independence, corruption, international connections, and facilities.

35. History

The simulation records major civilization history: reforms, projects, crises, wars, discoveries, political changes, economic transformations, and other milestones.

History is a first-class system and should let players understand how their civilization developed.

36. Terminal Interface

Primary commands: overview, economy, population, politics, military, diplomacy, technology, resources, industries, infrastructure, regions, institutions, projects, intelligence, history.

Deep inspection commands should answer questions such as: why did food prices increase? why did unemployment rise? why is steel production falling? where is government money going? what industries depend on coal? what uses do horses currently have? why is research slow?

37. Strategic Paths

Industrialization: education → engineering → factories → railways → mass production → urbanization.

Commercial Empire: ports → merchant fleet → trade agreements → financial institutions → foreign investment → economic influence.

Military Power: industrial base → military institutions → officer corps → arms production → logistics → professional army.

Scientific Civilization: education → universities → research institutions → scientific community → industrial R&D; → technological leadership.

38. Failure States

Possible systemic failures include bankruptcy, civil war, revolution, military coup, economic collapse, famine, political fragmentation, foreign occupation, technological stagnation, administrative collapse, and separatism.

Failure should emerge from simulation rather than arbitrary game-over screens.

39. Multiplayer

Each player controls a civilization in a persistent shared world.

Players interact through diplomacy, trade, alliances, espionage, wars, technology exchange, migration, investment, treaties, sanctions, and international organizations.

40. Offline Civilization Behavior

While a player is offline, their civilization continues according to existing policies, institutional behavior, AI government priorities, economic processes, military orders, and active projects.

The player should return to a civilization that has genuinely continued evolving.

41. Server Architecture

Initial architecture: Player → Astro/React → Supabase → PostgreSQL → Simulation Worker.

Supabase Cron can trigger simulation processing, while Edge Functions execute relevant simulation logic. Netlify primarily serves the web application.

42. Tick Safety and Performance

Simulation execution must be deterministic where possible, idempotent, transactional, and recoverable.

The system must handle missed ticks, duplicate execution, server failure, partial execution, and concurrent execution without corrupting world state.

Do not simulate every citizen individually. Primary simulation entities are population cohorts, businesses, industries, markets, political groups, military units, institutions, and regions.

43. Development Roadmap

Phase 1 — Simulation Core: world, regions, resources, population, cities, basic economy, time, ticks.

Phase 2 — Economy: goods, production, consumption, businesses, markets, prices, wages, taxation, government budget.

Phase 3 — Civilization Systems: institutions, infrastructure, technology, administrative capacity, political groups, laws.

Phase 4 — Diplomacy & Military: armies, logistics, war, diplomacy, treaties, trade.

Phase 5 — AI: AI civilizations, political actors, business behavior, military AI, diplomatic AI.

Phase 6 — Multiplayer: player civilizations, authentication, realtime updates, diplomacy, player interaction, persistent world.

44. Core Gameplay Loop

Observe → Understand → Plan → Decide → Invest / Legislate / Order → Simulation → Consequences → New Situation → Observe again.

The player is not primarily asking 'What technology should I unlock?' They are asking: **What kind of civilization am I trying to build, and what needs to change for that civilization to actually become possible?**

45. Core Differentiator

Traditional 4X: I have horses → I unlock cavalry.

This game: A region develops horse breeding; surplus horses lower prices; saddlery becomes profitable; roads improve; farmers adopt animal transport; the army develops mounted reconnaissance; the government creates postal routes; communication improves; regional commerce expands; horse culture develops; the region becomes internationally known for breeding.

This systemic chain is the heart of the game.

Appendix — Design Rules

- Never treat an important resource as a simple boolean unlock.
- Prefer emergent capabilities over flat technology bonuses.
- Separate knowledge, techniques, institutions, infrastructure, and actual capability.
- The player controls policy, not reality.
- Actors should have incentives and interests independent of the player.
- Whenever possible, model causal chains so players can investigate why outcomes occurred.
- Geography should matter to production, trade, administration, and warfare.
- Technology should transform how a civilization functions.
- Institutions should accumulate quality, knowledge, reputation, and history.
- The simulation should continue meaningfully when players are offline.